

Satvik Kaul

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Professional Summary

MSc Computer Science student with hands-on experience building AI/ML systems and data-driven applications using Python, PyTorch, TensorFlow, and FastAPI. Strong foundation in machine learning, computer vision, and applied NLP, with experience taking projects from experimentation to evaluation and deployment. Known for clear communication, collaborative problem-solving, and delivering reliable work in fast-paced team environments.

Skills

- **Programming Languages :** Python, C/C++, JavaScript, React, Java, SQL
- **AI/ML:** Machine Learning, Computer Vision, Natural Language Processing, TensorFlow, Pytorch, Scikit-learn, Model Evaluation
- **Software Development:** FastAPI, React Native, Node.js, REST APIs, PostgreSQL, MongoDB, Data Structures & Algorithms
- **Tools & Platforms:** Git, GitHub Actions, Docker, Agile/Scrum, Sharepoint, Workflow Automation

Professional Experience

Toronto Metropolitan University

February 2026 - Present

Graduate Researcher

Toronto, ON, Canada

- Develop and extend a self-supervised learning pipeline on the SICAPv2 dataset for 4-class Gleason grading, benchmarking CAE pretext learning and contrastive SSL (SimCLR; currently extending to MoCo v2 / BYOL) using a shared encoder, linear evaluation, and end-to-end fine-tuning, with an ablation plan covering SSL objective, and fine-tuning strategy.
- Current results (to date): CAE achieves 62.8% downstream accuracy and Cohen's $k = 0.500$, outperforming the supervised baseline (47.6%, $k = 0.341$); the current SimCLR baseline reaches 37.6% with $k = 0.067$, and results are tracked and compared under cross-fold validation.

Teaching Assistant

September 2025 - Present

- Supported labs, tutorials, and office hours for introductory programming courses, helping students build strong foundations in C, Python, debugging, and algorithmic problem-solving.
- Graded weekly labs and assessments with rubric-aligned feedback, improving clarity around code quality, correctness, and problem-solving technique.
- Developed simple marking scripts and sample solutions, and collaborated with instructors to maintain consistent grading standards and timely assessment turnaround.

Ministry of the Attorney General

January 2024 - October 2024

IT Analyst

Toronto, ON

- Led a 3-person Agile team to develop and deploy a custom ticketing portal using Microsoft Power Apps and Visio · automation, streamlining manual workflows by 25%, cutting input errors by 15%, and improving response time by 20%
- Resolved recurring MS Teams and Outlook issues during live hearings and operations, reducing helpdesk escalations · by ~30% and improving session reliability across 10+ hearing rooms
- Developed and maintained 10+ training guides for new AV and M365 tools, enabling rapid onboarding and reference · for adjudicators and tribunal staff.

System Support Officer (Co-op)

May 2021 - August 2021

- Contributed to the launch of the Ontario Land Tribunal's revamped website and intranet using Linux/Unix OS and · WordPress, improving content accessibility for 5,000+ monthly users and streamlining internal document access

- Developed a SharePoint-based workflow for the Accounts department, enhancing document and task tracking, resulting in a 30% reduction in task completion delays and improved communication skills across the team.
- Conducted user research and compiled a database of all 444 Ontario municipalities in Excel, supporting cross-functional project planning and external communications across departments.

Pizza Nova Ltd

August 2019 - December 2023

Team leader

North York, ON

- Trained and supervised a 20+ member team in a fast-paced, high-volume environment, ensuring consistent adherence to service protocols and improving team efficiency
- Designed and delivered weekly staff training on digital ordering systems and service workflows, resulting in an 18% boost in team productivity and reduced order handling time.

Projects & Outside Experience

Recipe Recommender System

Toronto, ON, Canada

- Designed a multi-modal pipeline: EfficientNet-B2 food classifier + Two-Tower recommender with GRU sequence modeling (ingredients + nutrition features; 1.1M Food.com interactions). Added context-aware image anchors and calorie-budget re-ranking; served via FastAPI/TypeScript with end-to-end latency <300 ms
- Results: NDCG@10 = 0.035 (3.6× over popularity; +23.7% vs. Matrix Factorization), HR@10 = 0.267, MRR = 0.108, Coverage = 34.2%; image classifier Top-1 50–65%, Top-5 85–90%
- [Link to project](#)

Java URL Shortener

Toronto, ON, Canada

- Built a URL shortening service in Java, incorporating RESTful API endpoints to create short links and manage user redirects with unique Base62-encoded cryptographic keys, while developing prototypes for future enhancements.
- Added strict URL validation (format/length/allowlist), lightweight rate limiting, request logging, and asynchronous redirect auditing to keep the service responsive and debuggable.
- [Link to project](#)

Education

Toronto Metropolitan University

September 2025 - August 2027

Master's, Computer Science

GPA: 4.22

Centennial College

September 2019 - June 2023

Bachelor's, Information Technology

GPA: 3.8